MAX PLANCK INSTITUTE
OF ANIMAL BEHAVIOR

ECOLOGY of
ANIMAL
SOCIETIES

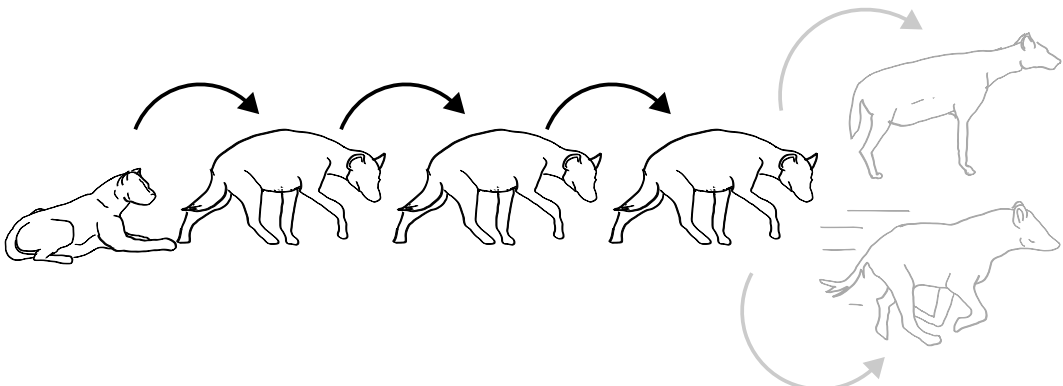
Communication
& Coordination
Across Scales

The Behavioural Observatory

Cross-species comparison of behavioural sequences in the wild

Pranav Minasandra

When to change your behavioural state?



When to change your behavioural state?

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- What does the algorithm that generates this set of behaviours look like?

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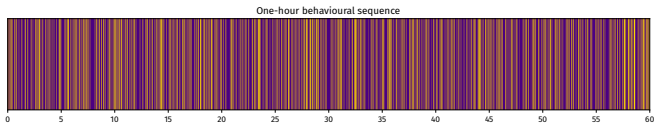

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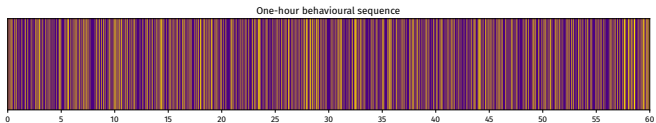
How an animal decides behaviour in the next instant.

- E.g., actions are chosen randomly each instant.
- Current-state-dependent transition probabilities



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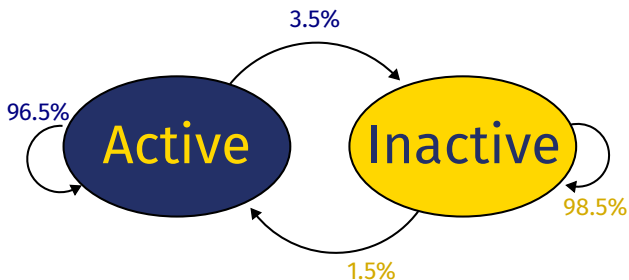

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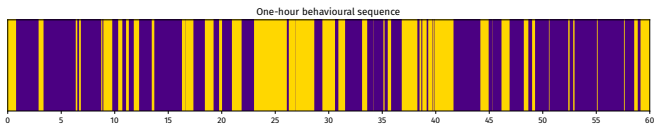
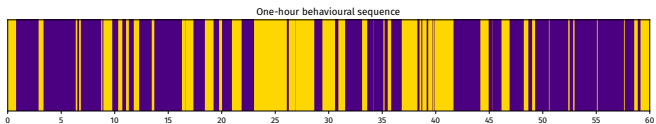


What is a behavioural algorithm?

How an animal decides behaviour in the next instant.

- E.g., actions are chosen randomly each instant.
- Current-state-dependent transition probabilities, i.e., a Markov model.
- Or something more complex.



Observing natural behaviour with accelerometers

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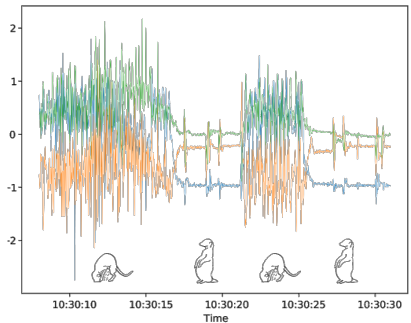

Observing natural behaviour with accelerometers

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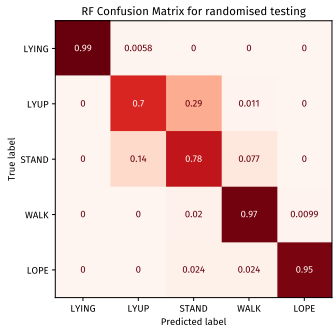
- Recorded using accelerometer collars.
- Behavioural sequences inferred using machine learning methods.



Exploring behavioural sequences

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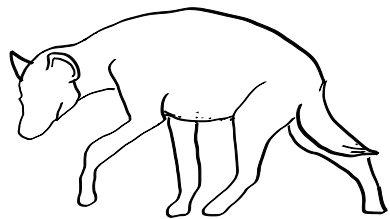
- A hyena has just started to walk.



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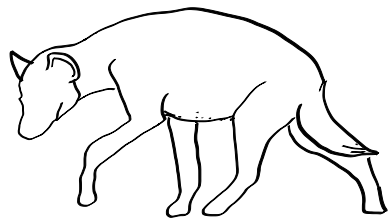

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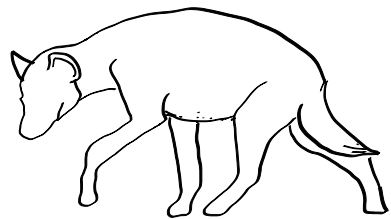

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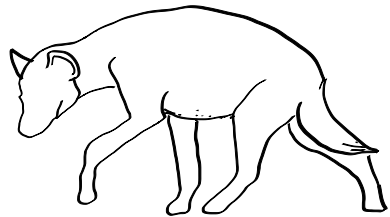

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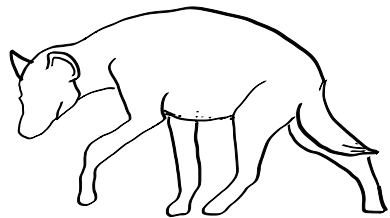

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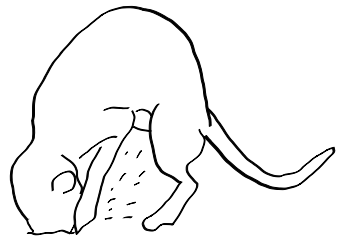

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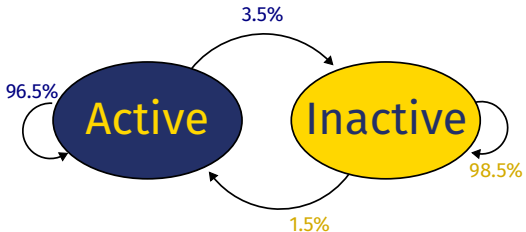

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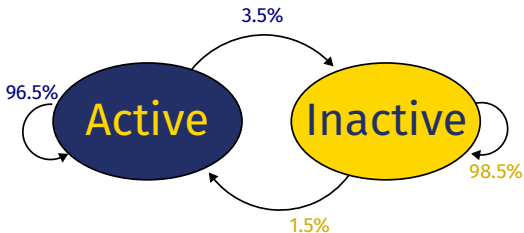

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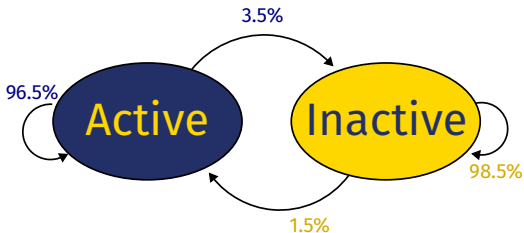
Hazard functions



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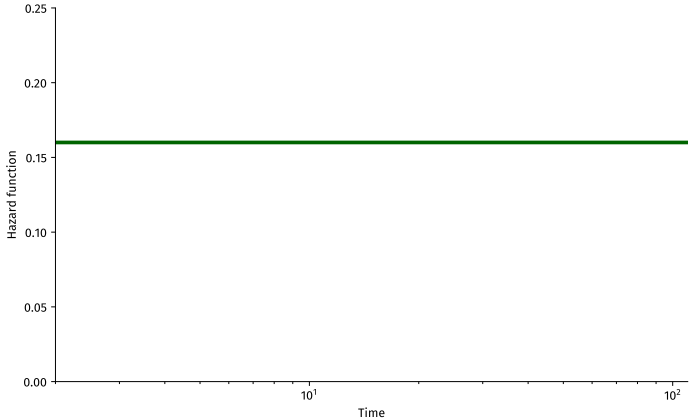

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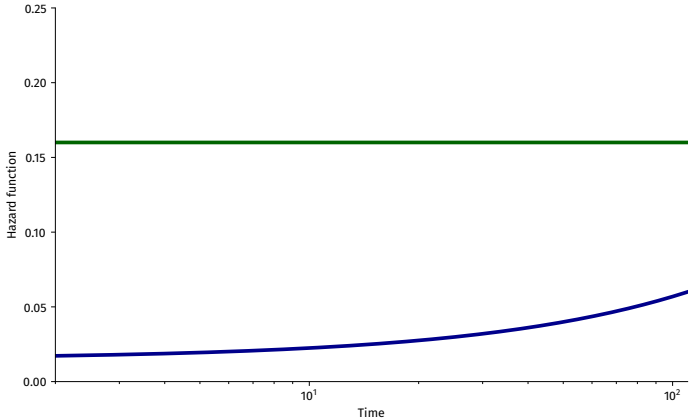

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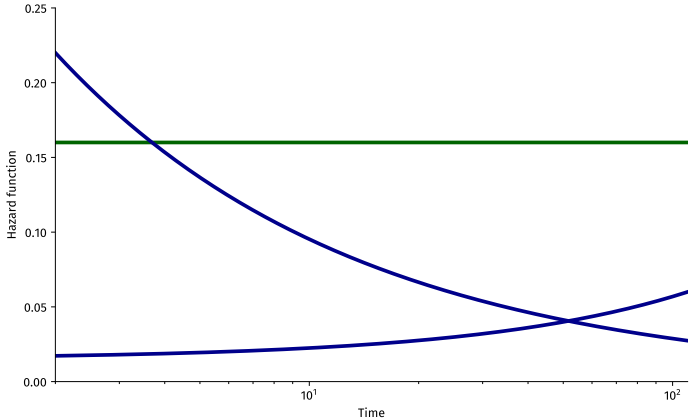
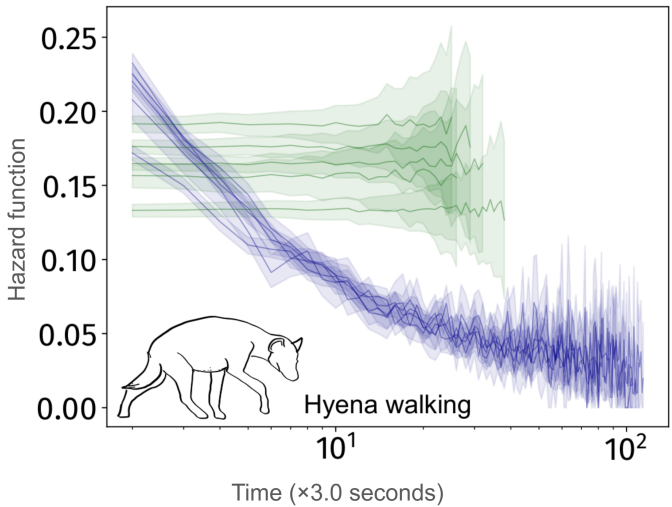


Hazard Functions—In reality



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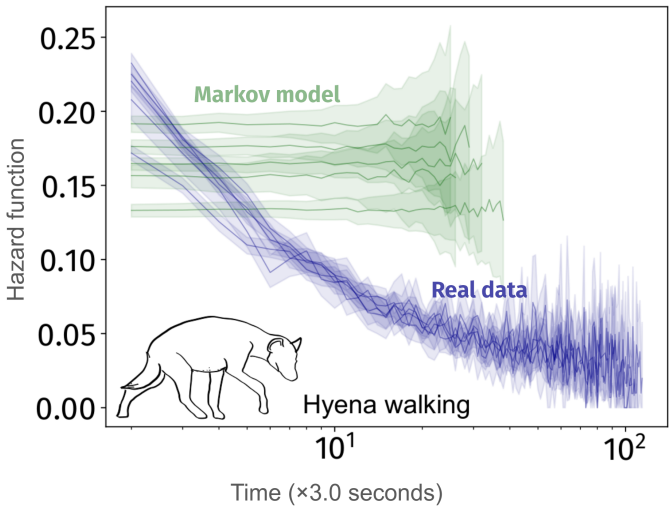

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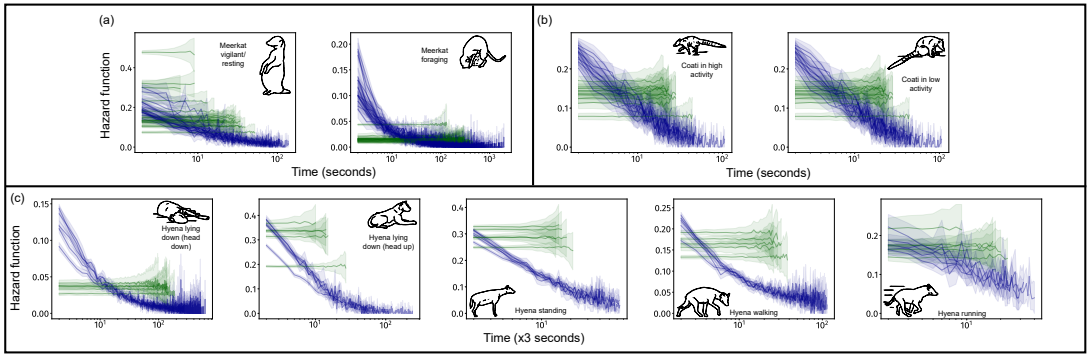

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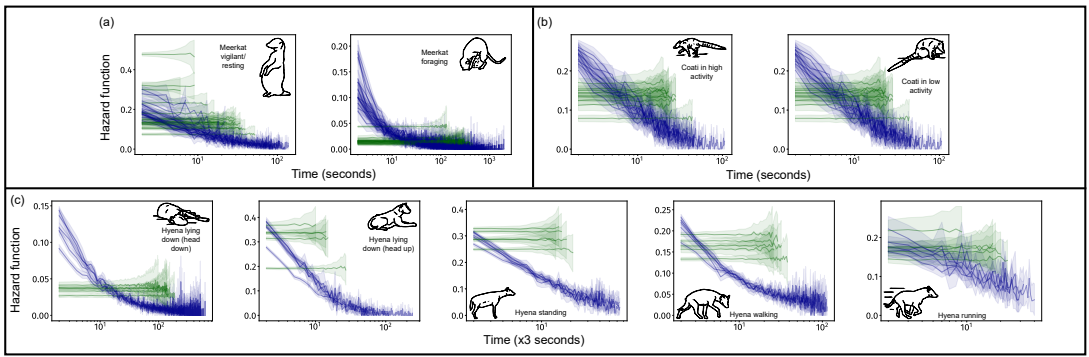

The longer an animal does something, the less likely it is to switch to do something else.

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Behavioral sequences across multiple animal species in the wild share common structural features

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Timescales of behaviour

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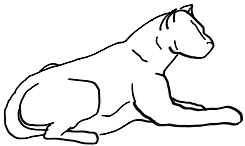

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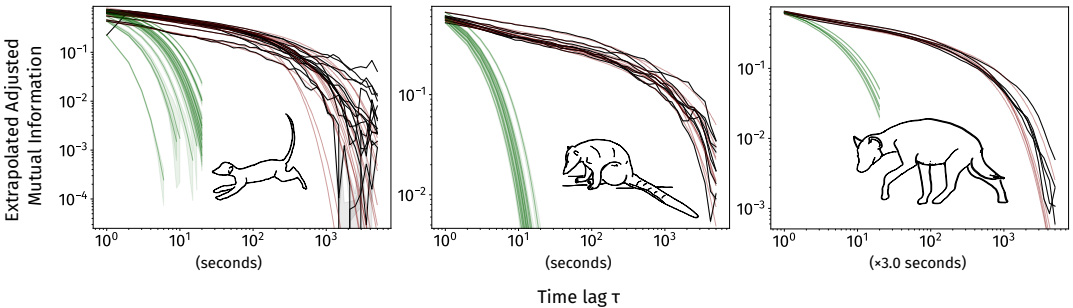

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Two of three shared results across species.

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- Are these universal phenomena?

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- Are these universal phenomena?
- Why do they happen?

Possibly quite widespread

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Cross-species behaviour delineation system

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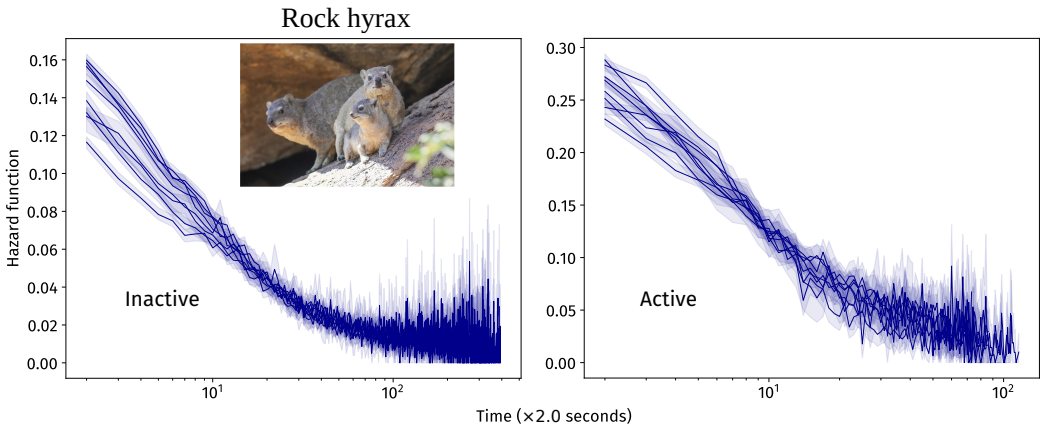

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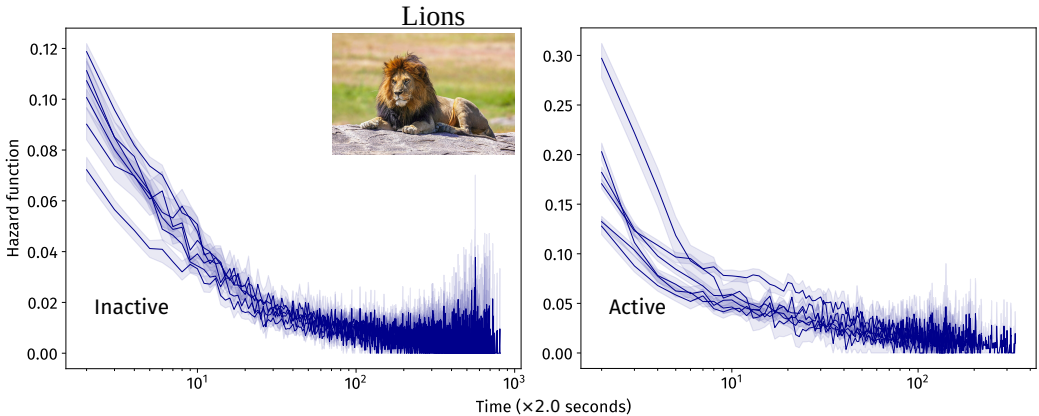

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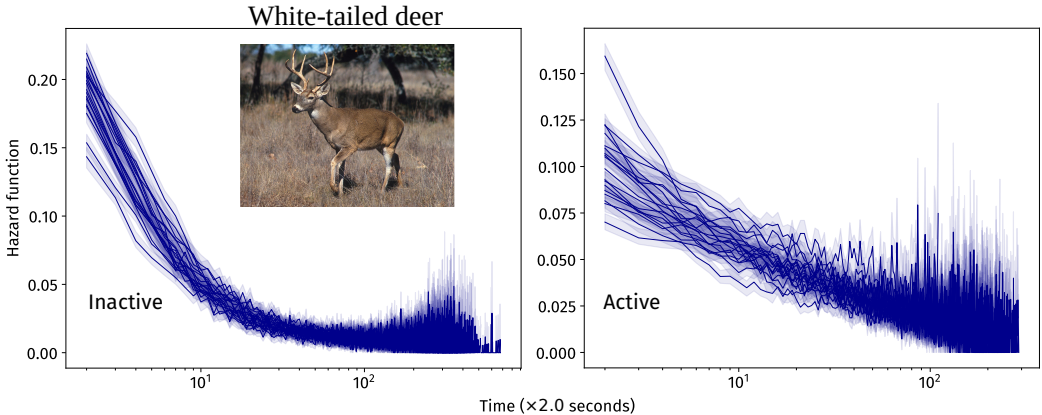
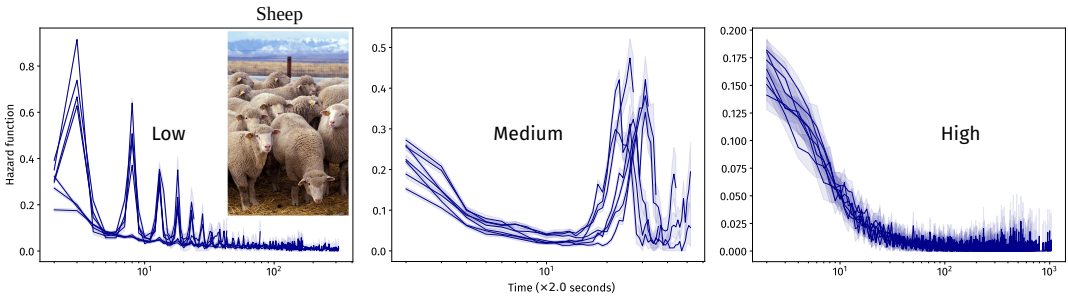


But what if all this is an artefact of the methods used?

Possibly quite widespread



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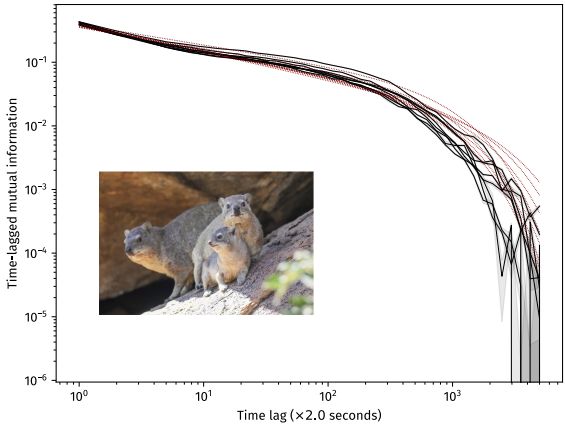

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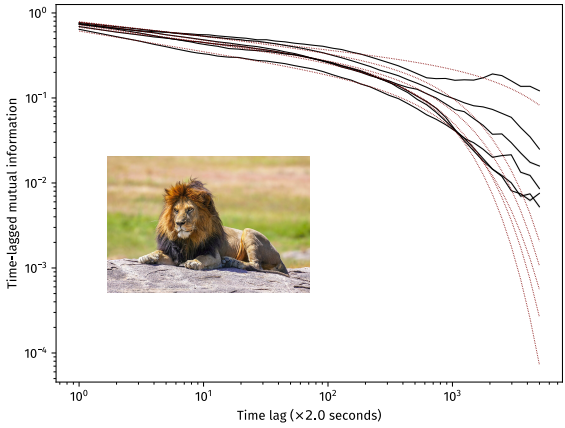

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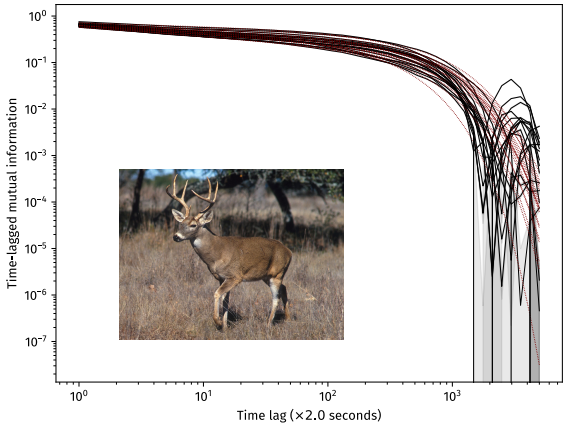

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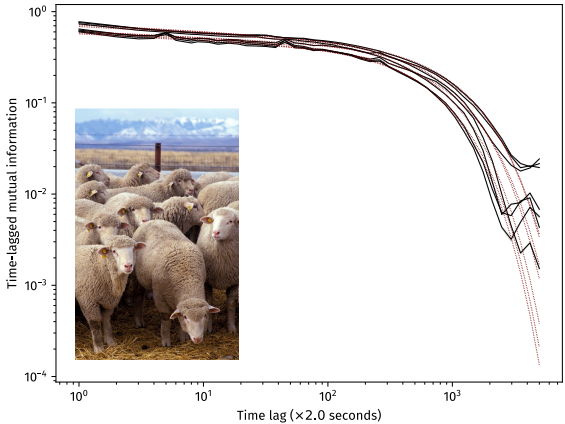

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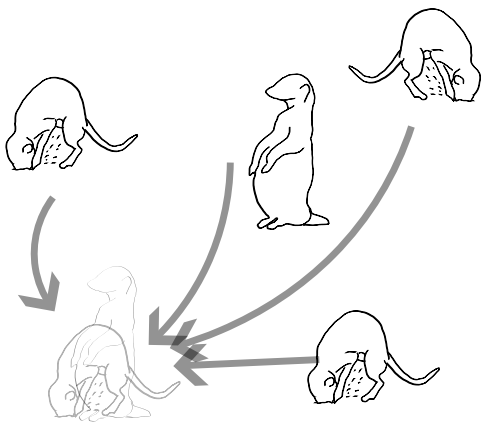

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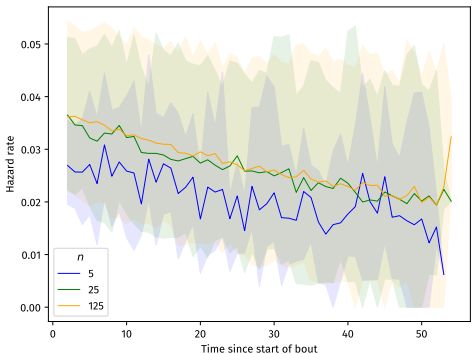

- Interacting animals tend to produce the same result.

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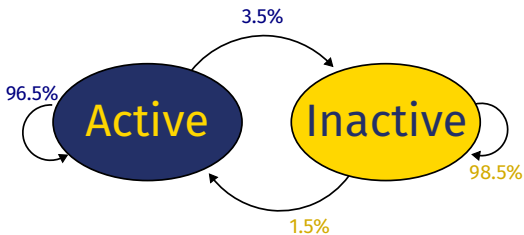

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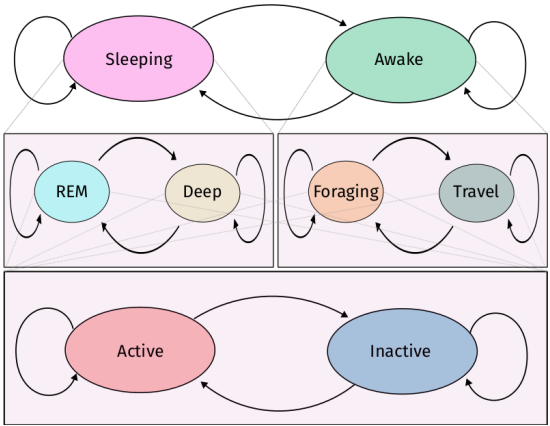

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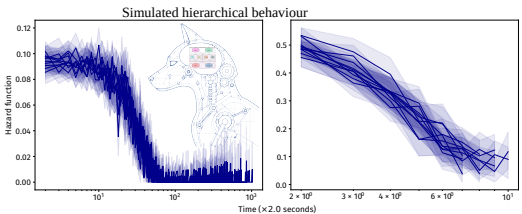
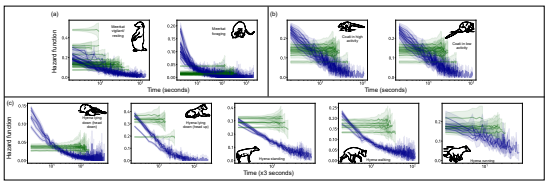


- Interacting animals tend to produce the same result.
- Hierarchical behaviour tends to produce the same result.

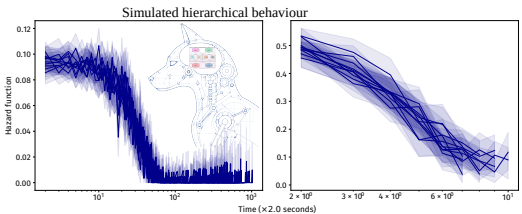
Take home messages

Take home messages



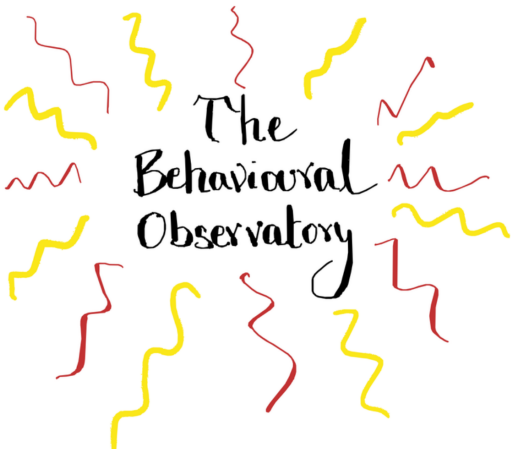
- A quite widespread behavioural algorithm.

Take home messages



- A quite widespread behavioural algorithm.
- Simulations allow us to test candidate behavioural algorithms.

Take home messages



- A quite widespread behavioural algorithm.
- Simulations allow us to test candidate behavioural algorithms.
- We welcome new collaborations.

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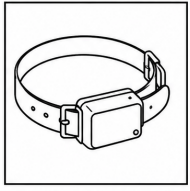
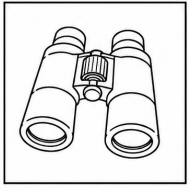
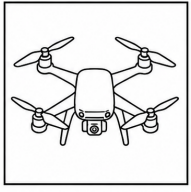
[This list is incomplete. You can help by expanding it.]

The Behavioural Observatory—Collaborators needed!







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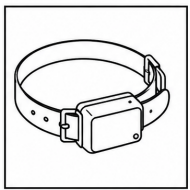

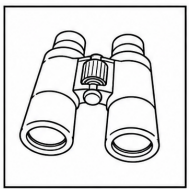

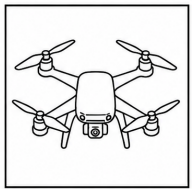

- Behavioural recordings that are continuous

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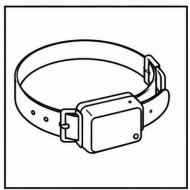
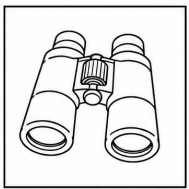
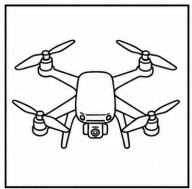

- Behavioural recordings that are continuous, multi-day,

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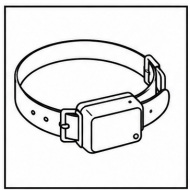
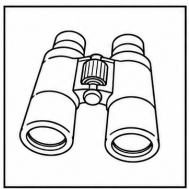
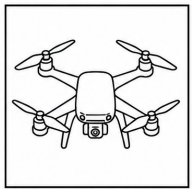
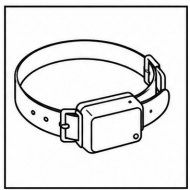
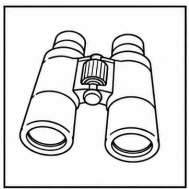
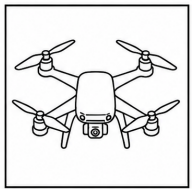




- Behavioural recordings that are continuous, multi-day, multi-individual.

The Behavioural Observatory—Collaborators needed!

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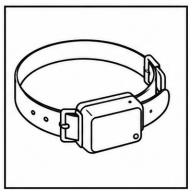
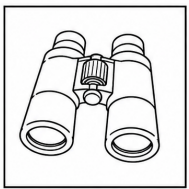
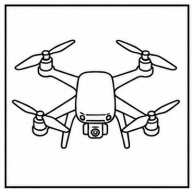

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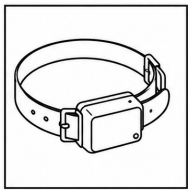

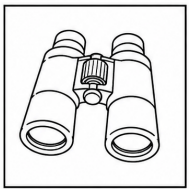

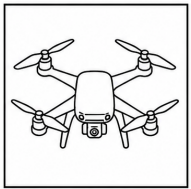

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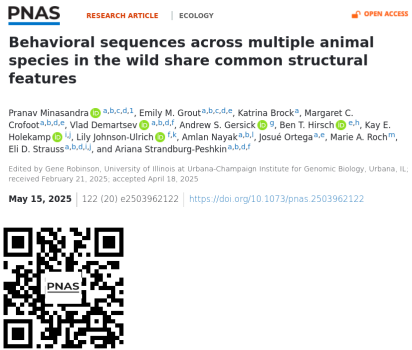

PNAS RESEARCH ARTICLE | ECOLOGY OPEN ACCESS

Behavioral sequences across multiple animal species in the wild share common structural features

Pranav Minasandra[a,b,c,d,1], Emily M. Grout[a,b,c,d,e], Katrina Brock[a], Margaret C. Crofoot[a,b,d,e], Vlad Demartsev[a,b,d,f], Andrew S. Gersick[g], Ben T. Hirsch[e,h], Kay E. Holekamp[i,j], Lily Johnson-Ulrich[f,k], Amlan Nayak[a,b,l], Josué Ortega[a,e], Marie A. Roch[m], Eli D. Strauss[a,b,d,l,j], and Ariana Strandburg-Peshkin[a,b,d,f]

Edited by Gene Robinson, University of Illinois at Urbana-Champaign Institute for Genomic Biology, Urbana, IL; received February 21, 2025; accepted April 18, 2025

May 15, 2025 | 122 (20) e2503962122 | https://doi.org/10.1073/pnas.2503962122

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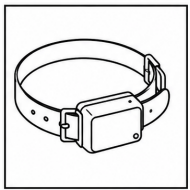

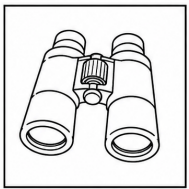

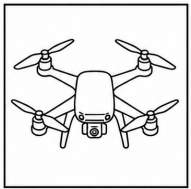

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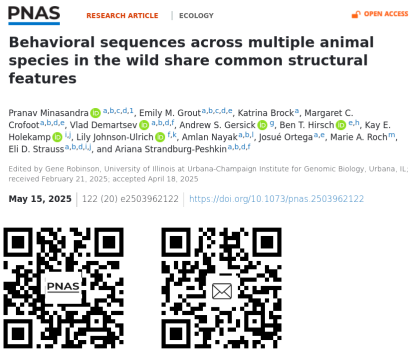

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